

GUMP

A Minimal GIMP Application:

Asa Levine

Content

1 Application



2 Function



3 Development



4 Management

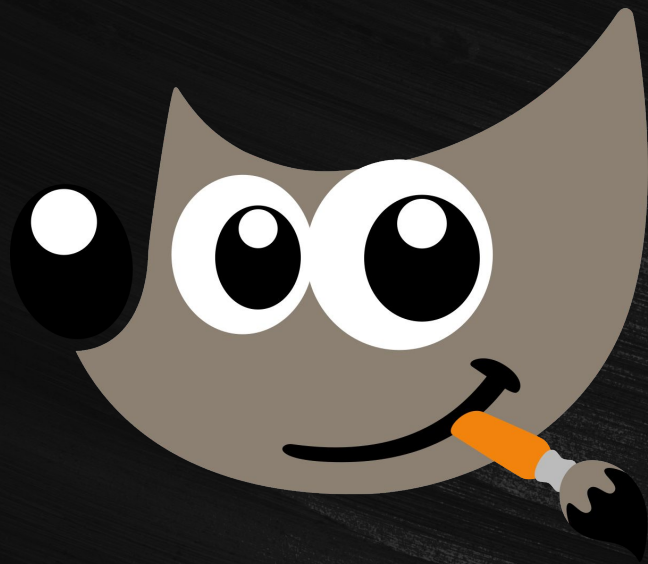


5 Status



6 Conclusion





application

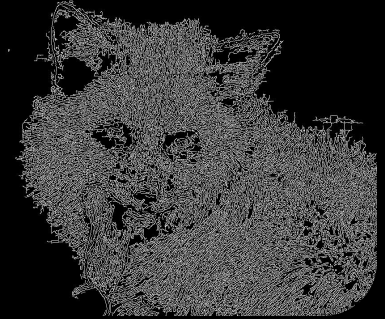
GIMP: GNU Image Manipulation Program

A free open-source raster image editing program released in 1995 and developed by Spencer Kimball and Peter Mattis

GUMP

The GUMP application provides a lightweight and efficient spread of image manipulation and detection algorithms.

It utilizes the OpenCV library and is developed in the C++ language, compiled by GCC, and maintained solely by Asa Levine



Operations:

Image Manipulation

- Erode/Dilate
- Lighten/Darken
- Resize
- Flip
- Stitch
- Canny Edge Detection
- Face Detection

Image Workflow

- Restore
- Save
- Undo/Redo
- Exit
- History

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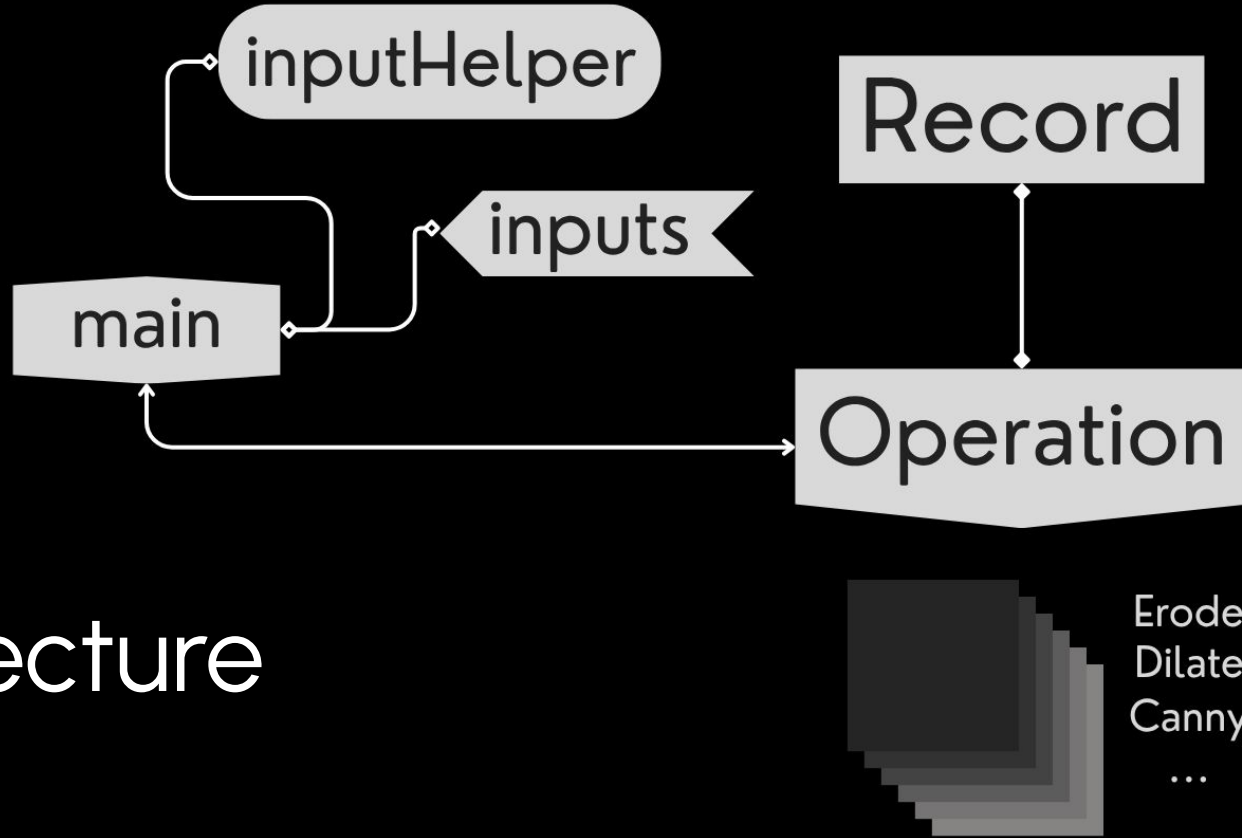
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Architecture

OpenCV

Available Functions

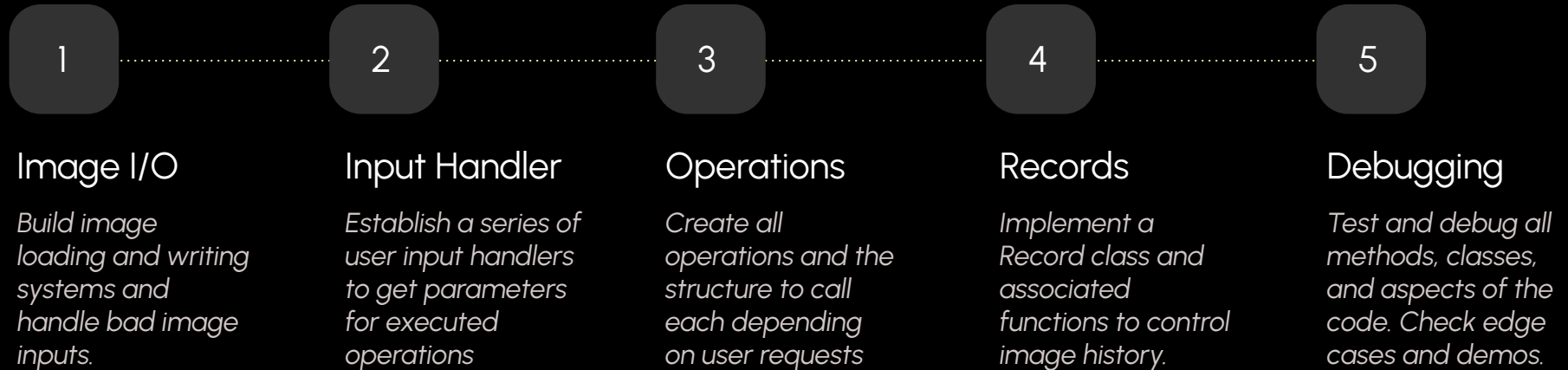
- Image reading/writing
- Video reading/writing
- image/video analysis
- Camera calibration
- 3D reconstruction
- Stitching
- Graphing
- Feature extraction
- Filtering
- Optical flow
- Frame transformations
- Object detection
- Face analysis
- Video stabilization
- 2D & 3D visualizers



OpenCV is the largest publicly available computer vision library, offering solutions both as both open-source and non-profit

<https://opencv.org>

Process



Workflow

Architecture Planning



Layout and plan all code architecture. This involves base systems, expanded features, and the library setup for OpenCV.

Development



Develop the full collection of code. This takes up majority of the time, and involves the integration of OpenCV into the GUMP application framework to allow for image manipulations.

Testing



Run QA tests to assure both positive user experience and catch any bugs not seen during the prior development portions.

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Current Functionality

Image Editing

A variety of different image editing functions are available to the user. These range from brightness control to edge detection, and all remain fully functional within their parameters.

Image Stitching

Both Scan and Panorama image stitching are possible with GUMP, using the initially uploaded image on top of further prompted images. It does not support image division during stitching.

History Control

A Record class allows for control over the image version. Undoing and redoing the operations is allowed for a default value of 100 images. This applies to all operations except for restore.

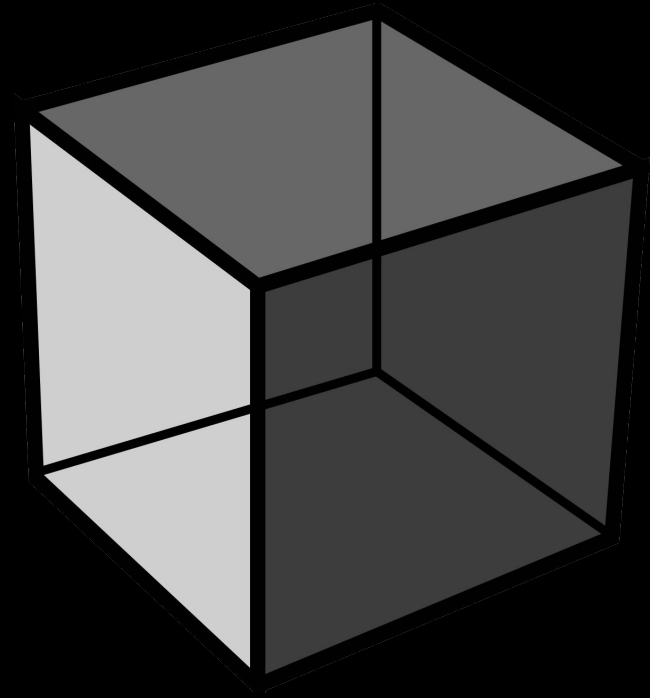
Face Detection

Simple face detection – highlighting the faces in red rectangles – is implemented without parameters for the user. It utilizes a pre-trained Yunanet face detection model within GUMP.

Known Issues

The local setup for OpenCV was partially flawed due to a deeper software issue on the development machine.

This resulted in GTK being uninitializable, stopping OpenCV's *imshow()* function from actually displaying any image, as the X11 framework is unable to actually initialize the window.



Looking Back

Strengths:

- Lightweight
- Efficient
- Approachable
- Minimal setup
- Version Control
- Modularity

Weaknesses:

- Limited
Parameterization
- No GUI
- Limited Face
Detection Control

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GUI



Design, test, and implement a dynamic and clean user interface to remove the bulkier and unclear cmd line visuals for users.

Mouse Events



Add mouse events to allow for shaped crops, drawing, and more for the user.

Mostly Requires the implementation of a GUI for proper usage.

Record+



Improve the Record system with better visibility and a more efficient storage method for past and future image retention.

File Types



Add support for more file types beyond those simply supported by OpenCV images – ex: videos, gifs, svgs, psds, and more.

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Thank You